Summary

Shadows decide where a Mars mission gets power, where the ground stays coldest, and what cameras can see, so TerraSync shows them next to height and slope. On our two measured HiRISE patches east of Jezero crater, a Sun 5° above the horizon puts on average about half of the checkable ground in shadow (51% on patch A, 46% on patch B). Raise the Sun to 30° and the shadow nearly disappears (under 1% on average).

The same ground can therefore look like an easy, sunny site or a dark, cold, hard-to-see site depending only on the Sun. An Earth analog that matches Mars on slope alone can still behave very differently once shadows are counted, which is why TerraSync compares shadows under the same Sun position for every site.

Where shadows change the study

Study area	What shadow changes	Why it matters for a mission
Solar power	A panel in shadow gets almost no direct sunlight.	Landers and rovers must be placed and parked where panels stay lit, especially in the low-Sun morning, evening and winter.
Surface temperature	Shaded ground loses heat without gaining it, so it stays colder.	Cold spots stress batteries and electronics, and are where frost and ice can survive longest.
Water ice and frost	Ice sublimates more slowly where it is cold and dark.	Shadowed pockets are targets when searching for ice and water resources.

Imaging and geology	Cameras see little detail inside dark shadows, but shadow edges reveal relief.	Shadow lengths are used to measure the height of rocks, cliffs and crater rims from a single image.
Rover driving	Stereo cameras struggle to judge rocks and holes hidden in shade.	Route planners prefer lit ground, or plan drives for a higher Sun.
Landing safety	Hazard cameras on a descending lander must see rocks and slopes.	Landing times are chosen so the target area is not dominated by long shadows.
Earth analog studies	Two sites with the same slopes can shade very differently, depending on how the slopes are arranged and which way they face.	A fair Earth-to-Mars comparison has to use the same Sun position for both sites.

The mathematics

Steps 1 to 6 are what the Mars pipeline (`Mars_Processed/pipeline.py`) computes for each 30 m cell. Steps 7 and 8 are standard follow-on calculations showing how shadow feeds into power and temperature; the site does not compute them yet.

1. Where the Sun is

The Sun is described by its altitude α (angle above the horizon) and azimuth β (clockwise from grid north). As a unit vector in East, North, Up coordinates:

$$\mathbf{s} = (\cos\alpha \sin\beta,\ \cos\alpha \cos\beta,\ \sin\alpha)$$

TerraSync uses 24 “what if” Sun positions: α = 5°, 15°, 30° and β = 0°, 45°, …, 315°.

2. Which way the ground faces

From the height grid z, the slopes in the east and north directions use central differences over two cells (about 60 m), with the true cell sizes Δx ≈ 29.91 m and Δy ≈ 29.99 m:

$$g_x = \frac{z_{i,j+1} - z_{i,j-1}}{2\,\Delta x}, \qquad g_y = \frac{z_{i-1,j} - z_{i+1,j}}{2\,\Delta y}$$

$$\theta = \arctan\sqrt{g_x^2 + g_y^2}, \qquad \mathbf{n} = \frac{(-g_x,\ -g_y,\ 1)}{\sqrt{1 + g_x^2 + g_y^2}}$$

Here θ is the slope and `n` is the unit vector pointing straight out of the ground.

3. How directly the light hits (cosine of incidence)

The incidence angle i is the angle between the sunlight and the ground's normal. Its cosine is the dot product `n · s`, which is exactly the formula in the pipeline:

$$\cos i = \frac{\sin\alpha - \cos\alpha\,(g_x \sin\beta + g_y \cos\beta)}{\sqrt{1 + g_x^2 + g_y^2}}$$

Written with slope θ and aspect A (the direction the slope faces, downhill):

$$\cos i = \cos\theta\sin\alpha + \sin\theta\cos\alpha\cos(\beta - A)$$

cos i = 1 means the Sun is straight overhead relative to that patch of ground; cos i = 0 means the light just grazes it. The Shadow map on the Mars page shades lit ground by this value.

4. Self-shadow: the ground faces away from the Sun

A cell is in self-shadow when it tilts away from the Sun:

$$\cos i \le 0 \iff \tan\theta\,\cos(\beta - A) \le -\tan\alpha$$

For a slope facing directly away from the Sun (β − A = 180°) this becomes cos i = sin(α − θ), so the cell is dark whenever its slope is steeper than the Sun is high (θ ≥ α). With the Sun at 5°, any

slope over 5° facing away from it is dark; at 30°, almost no slope on these patches is that steep.

5. Cast shadow: something else blocks the Sun

A cell can face the Sun and still be shaded by a hill between it and the Sun. The pipeline walks from each cell towards the Sun in steps of about 30 m, up to 1,200 m. The cell at height z_0 is in cast shadow if, at any distance d along that line, the terrain rises above the sunbeam:

$$z(d) - \frac{d^2}{2R} \;>\; z_0 + d\,\tan\alpha, \qquad 0 < d \le 1200\ \mathrm{m}$$

The term $d^2/(2R)$ lowers distant ground for the curvature of Mars (R = 3,396 km). At 1,200 m it is only 0.21 m. If the walk leaves the patch or hits missing data before finding a blocker, the cell is marked unresolved rather than lit.

The final state of each cell:

$$\mathrm{shadowed} = \mathrm{cast} \vee \mathrm{self}, \qquad \mathrm{lit} = \neg\mathrm{cast} \wedge \neg\mathrm{self}$$

6. Shadow length, and height from a shadow

An object of height h on flat ground casts a shadow of length L:

$$L = \frac{h}{\tan\alpha} \qquad \Longleftrightarrow \qquad h = L\,\tan\alpha$$

The second form is how planetary scientists measure the height of boulders, cliffs and crater rims from one orbital image when the Sun angle is known. For a 10 m tall object:

Sun altitude α	tan α	Shadow length of a 10 m object
5°	0.087	114 m
15°	0.268	37 m
30°	0.577	17 m

A Sun 6 times lower (5° vs 30°) makes shadows about 6.6 times longer. This is why low-Sun shadows cover so much more of the patches (section below).

7. Shadow fraction

The headline number for each Sun position counts only cells whose state is known:

$$f_{\text{shadow}} = \frac{N_{\text{shadowed}}}{N_{\text{shadowed}} + N_{\text{lit}}}$$

Unresolved cells (near the patch edge, where the walk ran out of map) are left out, so the share is not inflated or deflated by missing data.

8. Follow-on: power and temperature

Sunlight reaching Mars is weaker than at Earth because Mars is farther from the Sun (mean distance r ≈ 1.524 AU, solar constant at Earth S₀ ≈ 1,361 W/m²):

$$S_{\text{Mars}} = \frac{S_0}{r^2} \approx \frac{1361}{1.524^2} \approx 586 \ \text{W/m}^2$$

The direct sunlight on a piece of ground is zero in shadow and otherwise scaled by the incidence angle and by dust in the air (optical depth τ; the path through the air grows as 1/sin α):

$$E_{\text{direct}} = \begin{cases} S_{\text{Mars}}\, e^{-\tau/\sin\alpha}\, \cos i & \text{if lit} \\ 0 & \text{if shadowed} \end{cases}$$

Example with an assumed clear-day τ = 0.5 on flat ground: at α = 30° the direct beam is 586 × e^(−1) × 0.5 ≈ 108 W/m²; at α = 5° it is 586 × e^(−5.7) × 0.087 ≈ 0.2 W/m². A low Sun therefore costs power twice: weaker light and much more ground in shadow. Shaded ground still gets some scattered skylight, which this direct-beam formula leaves out.

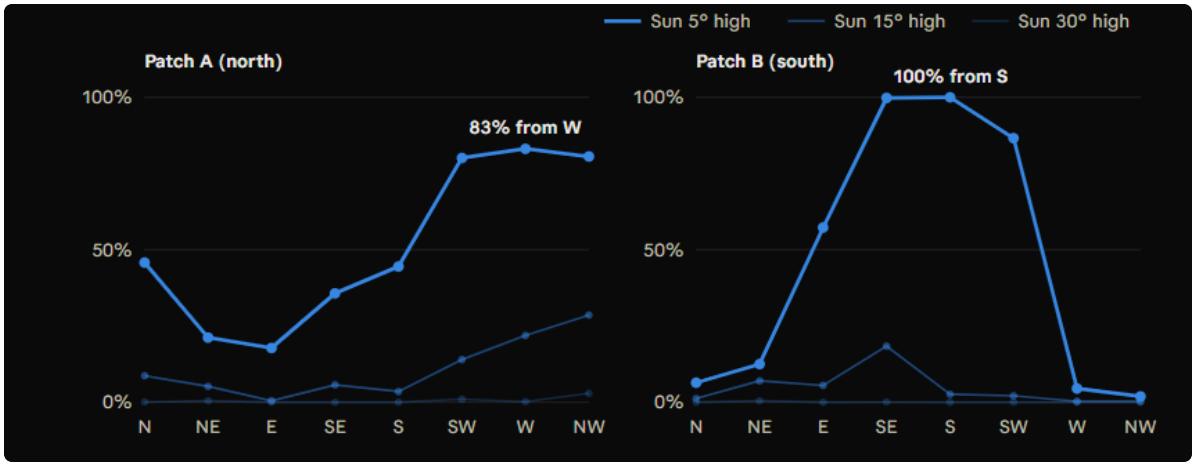
A rough upper limit on how warm sunlit ground can get comes from balancing absorbed sunlight against emitted heat (albedo a, emissivity ε, Stefan–Boltzmann constant σ = 5.67 × 10⁻⁸ W/m²K⁴):

$$T_{\max} \approx \left(\frac{(1-a)\,E}{\varepsilon\,\sigma} \right)^{1/4}$$

With a = 0.25, ε = 0.95 and the full 586 W/m² this gives about 300 K (27 °C). Ground in shadow has E ≈ 0 from the direct beam, so it can only cool, which is why shadows make the coldest spots on a site. Real temperatures also depend on how well the soil stores heat and on the thin atmosphere.

What the Mars patches show

The Sun’s height matters far more than its direction: averaged over the 8 directions, shadow drops from about half the ground at 5° to 5–11% at 15° and under 1% at 30°.



Direction still matters at a low Sun, and it is different for each patch. Patch A is darkest with the Sun in the west (83%), patch B with the Sun in the south (100%). Patch B is almost fully lit with a low Sun from the north-west (2%). The two patches have nearly the same average slope, so slope alone would have predicted the same shadows; only the full height map reveals the difference. That is the reason TerraSync shows shadow maps and not just slope numbers.

Limits and next steps

The shadow maps are geometry only, so they show where light is blocked, not how bright the ground looks to a camera.

- Hypothetical Sun. The 24 Sun positions are scenarios, not real dates or times. Next step: compute real Sun positions for a chosen season and hour at Jezero.
- 1,200 m search limit. Hills farther than 1,200 m are not checked, and cells whose search leaves the patch are unresolved (roughly 40–55% of cells at most Sun positions).
- 30 m cells. Rocks and small ridges smaller than a cell cast no shadow in this model, so real ground has more small shadows than shown.
- Grid north. Azimuth is measured from the map grid’s north, which has not been converted to true north.
- No atmosphere. Scattered skylight, dust and refraction are left out; shaded ground on Mars still receives some diffuse light.

- **Earth side.** To compare shadows with an Earth analog, the Earth site needs a height model (for example NASADEM, 30 m) processed with the same Sun positions, cell size and 1,200 m limit.